

Equation of motion

$$\bar{h}[k+1] = \bar{h}[k] + a_h[k]\{f_{dh}[k] + f_T[k]\} + \delta\bar{h}_D[k],$$

Motion error

$$\delta\bar{h}[k+1] = \delta\bar{h}[k] + \Delta\bar{h}_d[k] - a_h[k]\{f_{dh}[k] + f_T[k]\} - \delta\bar{h}_D[k],$$

$$\delta\bar{h}[k] = \bar{h}_d[k] - \bar{h}[k], \quad \Delta\bar{h}_d[k] = \bar{h}_d[k+1] - \bar{h}_d[k]$$

Measurement

(d -step delay)

$$\delta\bar{h}_m[k] = \delta\bar{h}[k-d] + n_h[k], \quad \textbf{Control objective : } \delta\bar{h}[k+1] = \lambda_c \delta\bar{h}[k], \quad 0 \leq \lambda_c < 1$$

Ideal control effort (no measurement delay), and the resulting motion error:

$$\bar{f}_{dh}[k] = a_h^{-1}[k]\{\Delta\bar{h}_d[k] + (1 - \lambda_c)\delta\bar{h}[k] - \delta\bar{h}_D[k]\}$$

$$\delta\bar{h}[k+1] = \delta\bar{h}[k] + \Delta\bar{h}_d[k] - a_h[k]\{\bar{f}_{dh}[k] + f_T[k]\} - \delta\bar{h}_D[k] = \lambda_c \cdot \delta\bar{h}[k] - a_h[k]f_T[k]$$

d-step measurement delay:

$$\delta\bar{h}[k] = \delta\bar{h}[k-d] + \sum_{i=1}^d \Delta\bar{h}_d[k-i] - \sum_{i=1}^d a_h[k-i]\{f_{dh}[k-i] + f_T[k-i]\} - \sum_{i=1}^d \delta\bar{h}_D[k-i]$$

Ideal control effort (with d-step measurement delay):

$$\begin{aligned} \bar{f}_{dh}[k] = a_h^{-1}[k] \Big\{ & \Delta\bar{h}_d[k] - \delta\bar{h}_D[k] + (1 - \lambda_c) \sum_{i=1}^d \left[\Delta\bar{h}_d[k-i] - \delta\bar{h}_D[k-i] \right] \\ & + (1 - \lambda_c) \left[\delta\bar{h}[k-d] - \sum_{i=1}^d a_h[k-i]f_{dh}[k-i] \right] \Big\} \end{aligned}$$

$$\delta\bar{h}[k+1] = \lambda_c \cdot \delta\bar{h}[k] - \left\{ a_h[k]f_T[k] + (1 - \lambda_c) \sum_{i=1}^d a_h[k-i]f_T[k-i] \right\}$$

Implementable control effort

(disturbance not estimated)

$$\boxed{f_{dh}[k] = \hat{a}_h^{-1}[k] \left\{ \Delta\bar{h}_d[k] + (1 - \lambda_c) \left[\sum_{i=1}^d \Delta\bar{h}_d[k-i] + \delta\bar{h}_m[k] - \sum_{i=1}^d \hat{a}_h[k-i]f_{dh}[k-i] \right] \right\}}$$

Motion error

$$\delta\bar{h}[k+1] = \lambda_c \cdot \delta\bar{h}[k] - a_h[k] [f_{dh}[k] - \bar{f}_{dh}[k]] - \left\{ a_h[k] f_T[k] + (1 - \lambda_c) \sum_{i=1}^d a_h[k-i] f_T[k-i] \right\}$$

$$\begin{aligned} a_h[k] f_{dh}[k] - a_h[k] \bar{f}_{dh}[k] &= e_{a_h}[k] f_{dh}[k] + (1 - \lambda_c) n_h[k] \\ &\quad + (1 - \lambda_c) \sum_{i=1}^d (a_h[k-i] - \hat{a}_h[k-i]) f_{dh}[k-i] \\ &\quad + \delta\bar{h}_D[k] + (1 - \lambda_c) \sum_{i=1}^d \delta\bar{h}_D[k-i] \end{aligned}$$

Gain-error back-step:

$$\begin{aligned} a_h[k-i] - \hat{a}_h[k-i] &\approx e_{a_h}[k] - (\bar{h}_d[k] - \bar{h}_d[k-i]) (a'_h[k] - \hat{a}'_h[k]) \approx e_{a_h}[k] \\ a_h[k] f_{dh}[k] - a_h[k] \bar{f}_{dh}[k] &\approx e_{a_h}[k] \underbrace{\left\{ f_{dh}[k] + (1 - \lambda_c) \sum_{i=1}^d f_{dh}[k-i] \right\}}_{F_{dh}[k]} \\ &\quad + \delta\bar{h}_D[k] + (1 - \lambda_c) \sum_{i=1}^d \delta\bar{h}_D[k-i] + (1 - \lambda_c) n_h[k] \end{aligned}$$

$\begin{aligned} \delta\bar{h}[k+1] &= \lambda_c \cdot \delta\bar{h}[k] - F_{dh}[k] \cdot e_{a_h}[k] - \delta\bar{h}_D[k] - (1 - \lambda_c) \sum_{i=1}^d \delta\bar{h}_D[k-i] - \varepsilon_h[k] \\ \varepsilon_h[k] &:= a_h[k] f_T[k] + (1 - \lambda_c) \sum_{i=1}^d a_h[k-i] f_T[k-i] + (1 - \lambda_c) n_h[k] \end{aligned}$

Gain model

($a_o = \Delta t / (\gamma_N R)$ = far-field gain, $\bar{h}_d = h_d / R$; a_h has units 1/pN)

$$a_h(\bar{h}) = a_o [1 - e^{-b\varphi(\bar{h})}], \quad \varphi(1) = 0, \quad \varphi(\bar{h} \rightarrow \infty) = \infty, \quad \varphi' > 0$$

$$a'_h(\bar{h}) := \frac{da_h}{d\bar{h}} = \varphi'(\bar{h}) b a_o e^{-b\varphi(\bar{h})} = \varphi'(\bar{h}) b [a_o - a_h(\bar{h})]$$

$$B[k] := b \varphi'(\bar{h}_d[k]) \quad \implies \quad a'_h[k] = B[k] [a_o - a_h[k]] \quad (b \text{ is a state, } B[k] \text{ is not})$$

Shape function

$$\varphi(\bar{h}) = \ln \bar{h} \quad \implies \quad \varphi'(\bar{h}) = \frac{1}{\bar{h}}, \quad B[k] = \frac{b}{\bar{h}_d[k]}, \quad a_h(\bar{h}) = a_o [1 - \bar{h}^{-b}]$$

$a_h(\bar{h}) = a_o [1 - e^{-b\varphi(\bar{h})}],$	$a'_h[k] = B[k] [a_o - a_h[k]],$	$b[k+1] = b[k],$
$\varphi(\bar{h}) = \ln \bar{h},$	$B[k] = b \varphi'(\bar{h}_d[k]) = b / \bar{h}_d[k].$	

State

($\mathbf{y}$ carries $\delta\bar{h}[k-d]$; a_h and b are unknown)

$$\delta\bar{h}_j[k] := \delta\bar{h}[k-d+j-1], \quad j = 1, \dots, d+1; \quad \mathbf{x}[k] = [\delta\bar{h}_1, \delta\bar{h}_2, \delta\bar{h}_3, a_h, b]^\top[k] \quad (d=2)$$

Tracking error

(boxed motion error, $\delta\bar{h}_D \equiv 0$ hereafter)

$$\delta\bar{h}_3[k+1] = \lambda_c \delta\bar{h}_3[k] - F_{dh}[k] e_{a_h}[k] - \varepsilon_h[k]$$

Gain recursion

(the gain is evaluated at the true point, which trails the command by $\delta\bar{h}_3$)

$$\bar{h}[k] = \bar{h}_d[k] - \delta\bar{h}_3[k]$$

$$a_h[k] = a_h(\bar{h}_d[k]) - a'_h[k] \delta\bar{h}_3[k]$$

$$a_h[k+1] = a_h(\bar{h}_d[k+1]) - a'_h[k] \delta\bar{h}_3[k+1] = a_h(\bar{h}_d[k]) + a'_h[k] \Delta\bar{h}_d[k] - a'_h[k] \delta\bar{h}_3[k+1]$$

Subtracting,

$$a_h[k+1] = a_h[k] + a'_h[k] \underbrace{\left\{ \Delta\bar{h}_d[k] - (\delta\bar{h}_3[k+1] - \delta\bar{h}_3[k]) \right\}}_{\text{step taken by the true position}}$$

and substituting $\delta\bar{h}_3[k+1]$,

$$a_h[k+1] = a_h[k] + a'_h[k] \left\{ \Delta\bar{h}_d[k] + (1 - \lambda_c) \delta\bar{h}_3[k] + F_{dh}[k] e_{a_h}[k] + \varepsilon_h[k] \right\}$$

True state equation

$\begin{aligned} \delta\bar{h}_1[k+1] &= \delta\bar{h}_2[k] \\ \delta\bar{h}_2[k+1] &= \delta\bar{h}_3[k] \\ \delta\bar{h}_3[k+1] &= \lambda_c \delta\bar{h}_3[k] - F_{dh}[k] e_{a_h}[k] - \varepsilon_h[k] \\ a_h[k+1] &= a_h[k] + a'_h[k] \left\{ \Delta\bar{h}_d[k] + (1 - \lambda_c) \delta\bar{h}_3[k] + F_{dh}[k] e_{a_h}[k] + \varepsilon_h[k] \right\} \\ b[k+1] &= b[k], \quad a'_h[k] = B[k] [a_o - a_h[k]] \end{aligned}$
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Measurement

(every length, σ_{n_h} included, is already normalized by R)

$$\nabla_d \bar{h}_d[k] := \bar{h}_d[k] - \bar{h}_d[k-d]$$

$$y_1[k] = \delta\bar{h}_m[k] = \delta\bar{h}_1[k] + \underbrace{n_h[k]}_{r_1[k]}$$

$$y_2[k] = a_{hm}[k] = a_h[k-d] + \underbrace{n_a[k]}_{r_2[k]}$$

Delay back-off

$$a_h[k-d] = a_h[k] - a'_h[k] \nabla_d \bar{h}_d[k] \quad \implies \quad y_2[k] = a_h[k] - a'_h[k] \nabla_d \bar{h}_d[k] + n_a[k]$$

Linearization

(Jacobian at the estimate; partials at fixed a_h , since a_h and b are independent states)

$$\frac{\partial y_2}{\partial a_h} = 1 + \nabla_d \bar{h}_d[k] \hat{B}[k] \longrightarrow 1, \quad \frac{\partial y_2}{\partial b} = -\nabla_d \bar{h}_d[k] \frac{\hat{a}'_h[k]}{\hat{b}}$$

Output equation

($\mathbf{H}$ is the Jacobian; the second row is nonlinear in $\mathbf{x}$, so $\mathbf{H}\mathbf{x}$ is its linearization, not an identity)

$$\mathbf{y}[k] = \mathbf{H}[k] \mathbf{x}[k] + \mathbf{r}[k], \quad \mathbf{H}[k] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & -\nabla_d \bar{h}_d[k] \hat{a}'_h[k]/\hat{b} \end{bmatrix}, \quad \mathbf{r}[k] = \begin{bmatrix} n_h[k] \\ n_a[k] \end{bmatrix}$$

Estimation error

$$e_{\delta \bar{h}_j} = \delta \bar{h}_j - \hat{\delta \bar{h}}_j \quad (j = 1, 2, 3), \quad e_{a_h} = a_h - \hat{a}_h, \quad e_b = b - \hat{b}, \quad \mathbf{e} = [e_{\delta \bar{h}_1}, e_{\delta \bar{h}_2}, e_{\delta \bar{h}_3}, e_{a_h}, e_b]^\top$$

Innovation

($\hat{a}'_h = \hat{B}[k][a_o - \hat{a}_h]$, $\hat{B}[k] = \hat{b} \varphi'(\bar{h}_d[k])$; the second line mixes evaluation points, which is what makes it exact)

$$e_{y_1} = y_1 - \hat{\delta \bar{h}}_1 = e_{\delta \bar{h}_1} + n_h$$

$$a'_h - \hat{a}'_h = \varphi'(\bar{h}_d) \{b[a_o - a_h] - \hat{b}[a_o - \hat{a}_h]\} = -\hat{B}[k] e_{a_h} + \frac{a'_h[k]}{\hat{b}} e_b$$

$$\begin{aligned} e_{y_2} &= y_2 - (\hat{a}_h - \hat{a}'_h \nabla_d \bar{h}_d) = (1 + \nabla_d \bar{h}_d \hat{B}[k]) e_{a_h} - \nabla_d \bar{h}_d \frac{\hat{a}'_h[k]}{\hat{b}} e_b + n_a \\ &\longrightarrow e_{a_h} - \nabla_d \bar{h}_d \frac{\hat{a}'_h[k]}{\hat{b}} e_b + n_a \end{aligned}$$

Estimator

($\hat{a}_h$ swept by the step the estimator knows)

$$\hat{\delta \bar{h}}_1[k+1] = \hat{\delta \bar{h}}_2 + \ell_{11} e_{y_1} + \ell_{12} e_{y_2}$$

$$\hat{\delta \bar{h}}_2[k+1] = \hat{\delta \bar{h}}_3 + \ell_{21} e_{y_1} + \ell_{22} e_{y_2}$$

$$\hat{\delta \bar{h}}_3[k+1] = \lambda_c \hat{\delta \bar{h}}_3 + \ell_{31} e_{y_1} + \ell_{32} e_{y_2}$$

$$\hat{a}_h[k+1] = \hat{a}_h + \hat{a}'_h \{ \Delta \bar{h}_d + (1 - \lambda_c) \hat{\delta \bar{h}}_3 \} + \ell_{41} e_{y_1} + \ell_{42} e_{y_2}$$

$$\hat{b}[k+1] = \hat{b} + \ell_{51} e_{y_1} + \ell_{52} e_{y_2}$$

Error dynamics

(True – Estimator; coefficients at the estimate)

$$e_{\delta\bar{h}_1}[k+1] = e_{\delta\bar{h}_2} - \ell_{11}e_{y_1} - \ell_{12}e_{y_2}$$

$$e_{\delta\bar{h}_2}[k+1] = e_{\delta\bar{h}_3} - \ell_{21}e_{y_1} - \ell_{22}e_{y_2}$$

$$e_{\delta\bar{h}_3}[k+1] = \lambda_c e_{\delta\bar{h}_3} - F_{dh} e_{a_h} + \underbrace{(-\varepsilon_h)}_{q_3} - \ell_{31}e_{y_1} - \ell_{32}e_{y_2}$$

$$e_{a_h}[k+1] = \left(1 - [\Delta\bar{h}_d + (1-\lambda_c)\delta\hat{\bar{h}}_3]\hat{B} + \hat{a}'_h F_{dh}\right)e_{a_h} + (1-\lambda_c)\hat{a}'_h e_{\delta\bar{h}_3} + [\Delta\bar{h}_d + (1-\lambda_c)\delta\hat{\bar{h}}_3]\frac{\hat{a}'_h}{\hat{b}} e_b \\ + \underbrace{\hat{a}'_h \varepsilon_h}_{q_4} - \ell_{41}e_{y_1} - \ell_{42}e_{y_2}$$

$$e_b[k+1] = e_b - \ell_{51}e_{y_1} - \ell_{52}e_{y_2}$$

Error state equation

($\mathbf{q} = [0, 0, q_3, q_4, 0]^\top$, $q_3 = -\varepsilon_h$, $q_4 = \hat{a}'_h[k] \varepsilon_h = -\hat{a}'_h[k] q_3$; entries evaluated at the current estimate)

$$\mathbf{e}[k+1] = \mathbf{F}_e[k] \mathbf{e}[k] - \mathbf{L}[k] \mathbf{H}[k] \mathbf{e}[k] + \mathbf{q}[k]$$

$$\mathbf{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dh} & 0 \\ 0 & 0 & (1-\lambda_c)\hat{a}'_h & 1 - [\Delta\bar{h}_d + (1-\lambda_c)\delta\hat{\bar{h}}_3]\hat{B} + \hat{a}'_h F_{dh} & [\Delta\bar{h}_d + (1-\lambda_c)\delta\hat{\bar{h}}_3]\hat{a}'_h/\hat{b} \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Process noise Q

(only the current white thermal step enters Q : the past thermal steps in ε_h are cross-step correlated, and n_h already sits in R_1)

$$q_3 = -a_h[k] f_T[k], \quad q_4 = \hat{a}'_h[k] a_h[k] f_T[k] = -\hat{a}'_h[k] q_3$$

$$\gamma(\bar{h}) = \gamma_N c(\bar{h}), \quad \sigma_{f_T}^2 = \frac{4k_B T \gamma(\bar{h})}{\Delta t}, \quad a_h = \frac{\Delta t}{\gamma(\bar{h}) R}$$

$$\text{Var}(a_h f_T) = \hat{a}_h^2 \sigma_{f_T}^2 = \frac{4k_B T}{R} a_h[k]$$

$$\begin{aligned} Q_{33}[k] &= \frac{4k_B T}{R} a_h[k], & Q_{34}[k] &= Q_{43}[k] = -\hat{a}'_h[k] Q_{33}[k], \\ Q_{44}[k] &= \hat{a}_h'^2[k] Q_{33}[k], & Q_{55} &= 0 \quad (b \text{ constant}); \end{aligned}$$

All other entries vanish; the gain block is rank one, since $q_4 = -\hat{a}'_h[k] q_3$.

Measurement noise $R = \text{diag}(R_1, R_2)$

(a_{cov} = EWMA coefficient, IF_{var} = $\text{MA}(d)$ inflation factor of $\hat{\sigma}_{\delta\bar{h}_r}^2$)

$$R_1 = \sigma_{n_h}^2$$

$$\sigma_{\delta\bar{h}_r}^2[k] = C_{dpmr} \frac{4k_B T}{R} a_h[k] + C_n \sigma_{n_h}^2 = C_{dpmr} \frac{4k_B T}{R} (a_h[k] + \xi), \quad \xi := \frac{C_n}{C_{dpmr}} \frac{\sigma_{n_h}^2}{4k_B T/R}$$

$$a_{hm}[k] = \frac{\hat{\sigma}_{\delta\bar{h}_r}^2[k] - C_n \sigma_{n_h}^2}{C_{dpmr} (4k_B T/R)}$$

$$\text{Var}(\hat{\sigma}_{\delta\bar{h}_r}^2) = K_{var} \text{IF}_{var} (\sigma_{\delta\bar{h}_r}^2)^2, \quad K_{var} = \frac{2a_{cov}}{2 - a_{cov}}$$

$$\boxed{R_2[k] = K_{var} \text{IF}_{var} (a_h[k] + \xi)^2 + d Q_{44}[k]}$$

